Exam

Delft University of Technology – Faculty of Technology, Policy and Management

Course name: Introduction to TPM Modelling	Course code: EPA123A
Datum: **8th of April 2026**	**Time: 09:00 – 12:00**
Module manager: **Merla Kubli**	

Examination questions:
Number of questions: 10 questions
Max. number of points: **100** points
☐ **all questions have the same weight**
☒ **the questions have different weights** (indicated per question)

Total number of pages (incl. cover page): **12** pages

Use of tools and information sources:
During the examination, the use of tools or information sources is strictly forbidden unless stated below. This includes smartphones, smart watches, or any devices with similar functions.

Permitted tools and information sources for this exam:
☒ **books** ☒ **notes** ☒ **dictionaries** ☒ **readers** ☒ **formulae sheets**
☒ **calculator** ☒ **computer** ☐ **…**

All forms of communication are forbidden
Any suspicion of fraud or any breach of the exam rules will be immediately reported to the Board of Examiners. For more information about consequences of exam fraud: TU Delft Student portal> TPM> Rules and Guidelines.

Additional instructions:
- **Don't change the size of the boxes on the answer sheet**. Fit your answer in the boxes.
- Be extremely concise and clear when answering the open-answer questions.
- Put your name and student number on every file name you hand in.

To be handed in online on Brightspace:

☒ Examination answer sheet as a Word file named 'EPA123a_retake_student name_student number' as **.docx.**

Date when to expect the grades: 30th of April 2026

Honours pledge

0. (0 / 100) Regarding the exam of the course **EPA1324A** (Introduction to TPM Modelling), on April 8th, 2026, I declare that:

    a. I will take the exam on my own and within the allotted time.

    b. I will not have contact with any fellow student or other unauthorised people.

    c. I am aware that all rules regarding fraud at the TU Delft are applicable to this examination, and that fraud can lead to exclusion from the TU Delft.

    d. I have the necessary software (a well-functioning web browser, the correct version of MESA, Vensim Pro) correctly installed on my computer, and it is functioning as it should. It is my own responsibility to test these before the exam.

    Fill in "**YES**" on your answer form to confirm the statement above. If you choose to fill in "**NO**" or do not give an answer, you will not receive a grade for this exam.

ABM theory questions *(10 points)*

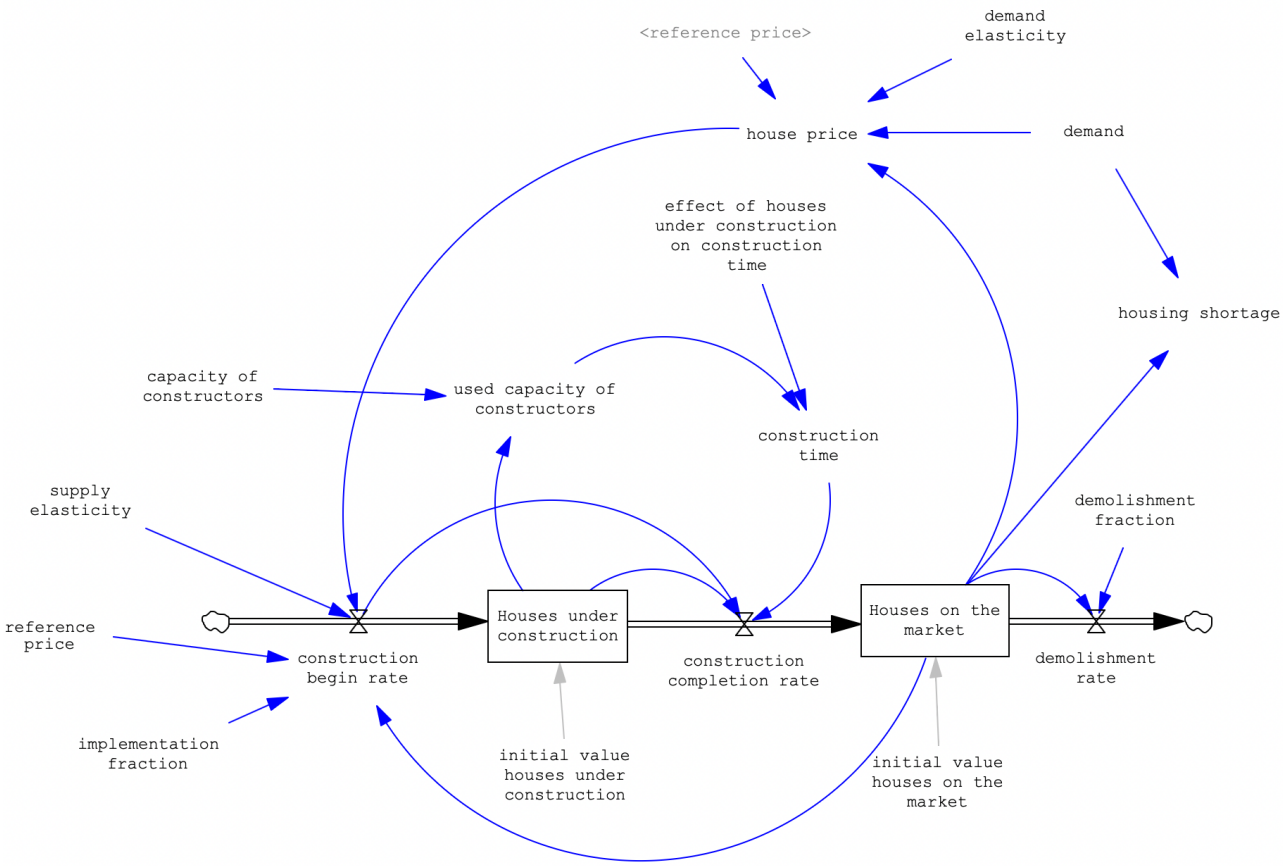
The war of the United States of America and Israel with Iran is causing massive disruptions to the flow of oil, LNG, fertilizer, and helium. Thousands of vessels are anchored in the Persian Gulf, waiting for the moment when they can safely sail through the Strait of Hormuz. This is causing unrest in the market for oil futures with knock on consequences for e.g., the price of gasoline or natural gas. At the same time, the sailing time from the strait to various destinations is in the order of 4-6 weeks, so physical supplies are only about to be affected. Moreover, both the emirates and Saudi Arabia have pipelines that bypass the strait, the capacity of which is being pushed to its physical maximum.

Imagine you are asked to develop a model to shed light on the dynamics of energy supply from the middle east for the remainder of this calendar year.

1. (4 / 100) What modelling method would you advise? Clearly argue why and explain the merits and drawbacks of the method you advise.
2. (3 / 100) Evidently, there is profound uncertainty about critical behavioural assumptions and exogenous input parameters. Name three critical uncertainties and explain how you would approach the handling of these uncertainties.
3. (3 / 100) How would you approach the validation of this model?

SD theory questions *(10 points)*

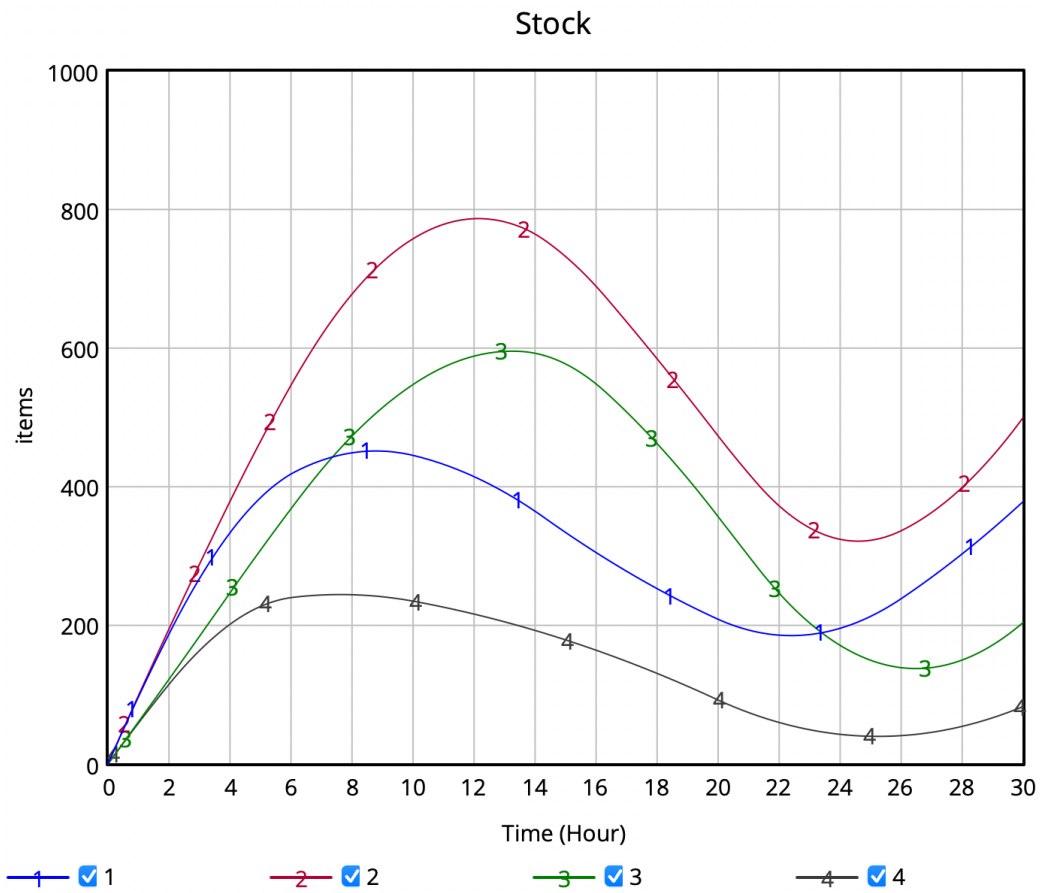
4. Consider the stock-flow diagram (SFD) on the housing shortage and the equations of all variables below.
    a. (3 / 100) Draw a disaggregated CLD of this system. Process all the information you have, including the equations. Higher-order delays may be represented in aggregated form. The elasticities may be assumed to be always elastic (larger than 1). (Note: Please draw the CLDs in Vensim and paste a screenshot.)
    b. (3 / 100) Draw an aggregated CLD of the same model.
    (Note: Please draw the CLDs in Vensim and paste a screenshot.)

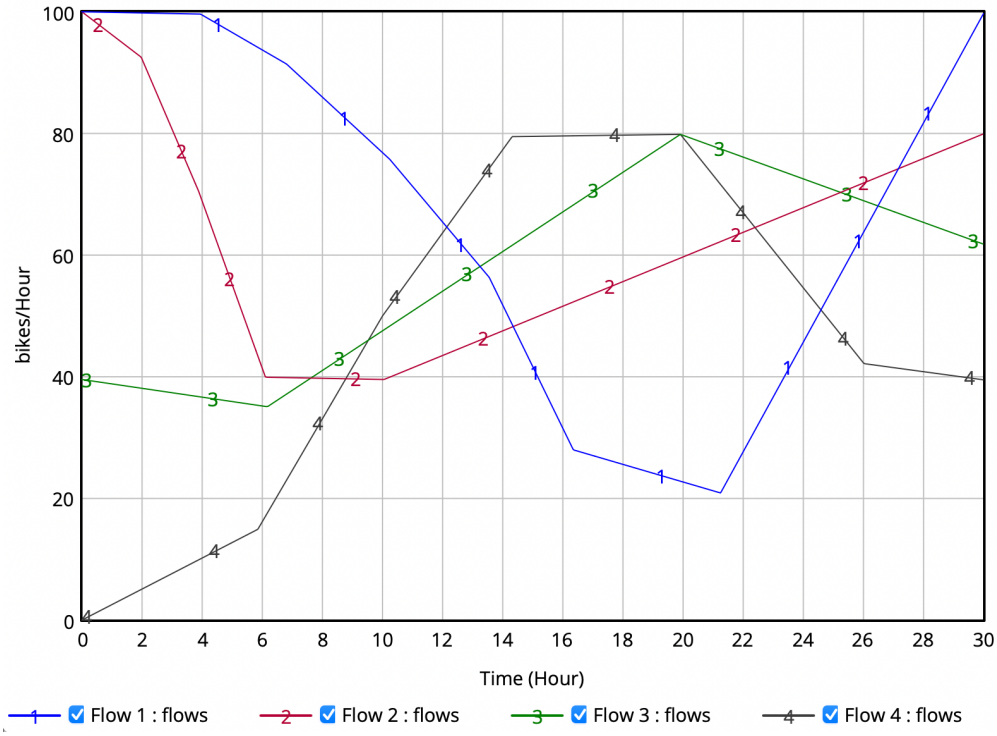


Model equations
housing shortage=demand-Houses on the market
Houses under construction= INTEG (construction begin rate-construction completion rate, initial value houses under construction)
Houses on the market= INTEG (construction completion rate-demolishment rate, initial value houses on the market)

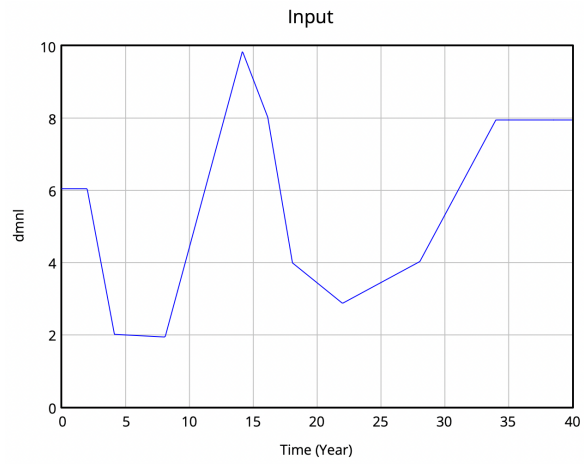
construction begin rate=(Houses on the market*(house price/reference price)^supply elasticity)*implementation fraction
construction completion rate=DELAY3I(construction begin rate, construction time, Houses under construction/construction time)
demolishment rate=Houses on the market*demolishment fraction
construction time=effect of houses under construction on construction time(used capacity of constructors)
used capacity of constructors=Houses under construction/capacity of constructors
effect of houses under construction on construction time=LOOKUP([(0,0)-(1,10)],(0,2), (0.149466,2.09), (0.311388,3.4), (0.441281,5.48611), (0.574733,7.70833), (0.80427,8.19444), (1,8), (1,8))
house price=reference price*(demand/Houses on the market)^demand elasticity
demand=2000
demand elasticity=1.05
demolishment fraction=1.03
capacity of constructors=18000
implementation fraction=0.9
initial value houses on the market=1000
initial value houses under construction=100
reference price=480000
supply elasticity=1.05

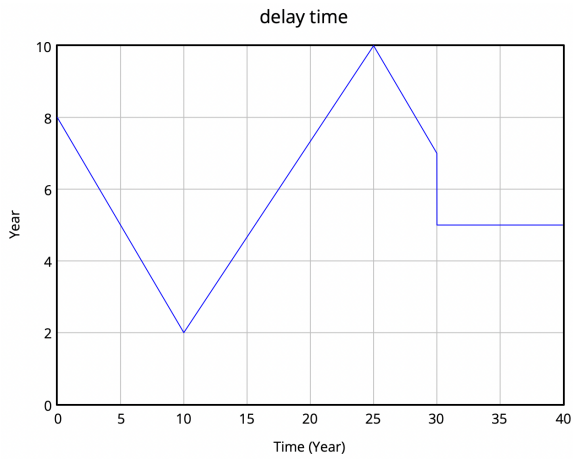
5. (2 / 100) Consider the following four stock behaviours and the stock's inflows (flow 1 and 2) and outflows (flow 3 and 4). Indicate for each run what the combinations of inflow and outflow result in the stock behaviour.

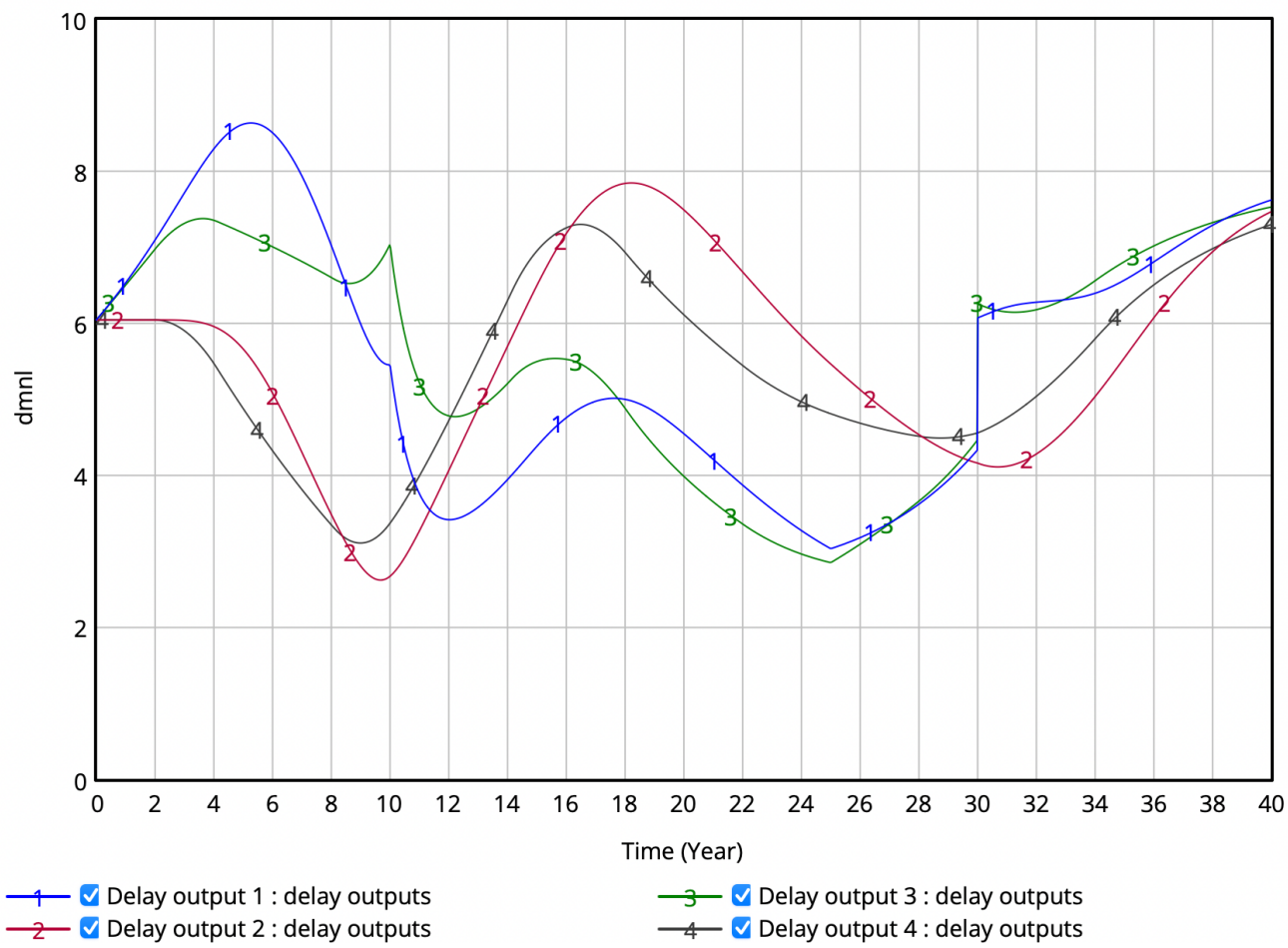




6. (2 / 100) Consider the figures below with four different delay outputs of the same input and delay time. What are the delay types (material or information) and orders (1st, 3rd, or FIXED) of the Delay outputs 1 to 4?







SD modelling questions (40 points)

Climate change mitigation

Climate change is one of the largest challenges for our society. The following model provides a high-level view of the decarbonization of the global energy system for the time period of 2015 to 2065. The description below provides insights into the model structure and data.

Sub-model of the Economy

The *GWP* ("Gross World Product" – which is a measure of the size of the global economy) is initially equal to 7.8e+013 dollar per year. *GWP* increases through the *Increasing GWP* variable, which is modelled as the difference between the *autonomous economic growth* of 3% per year and the *reduction of the GWP increase due to energy shortage*, times *GWP*. The *reduction of the GWP increase due to energy shortages* could, in turn, be modelled as the *fraction energy demand to energy supply* minus 1, times the *factor reduction of GWP increase due to energy shortages* of 10% per year.

Set the *energy intensity of GWP* initially equal to 5.54e-006 bbtu (billion British thermal unit) per dollar, and let it decrease further by means of a variable labelled *decreasing energy intensity of GWP*. This *decreasing energy intensity of GWP* variable could be modelled as the product of the *energy intensity of GWP* and the *investments in decoupling driven by energy shortages.* Model the *investments in decoupling driven by energy shortages* as the *decoupling due to energy shortages* of 25% per year times the third-order information delay of the positive part of the *fraction energy demand to energy supply* minus 1, delayed with an *investment delay time energy decoupling* of 8 years, and initial value equal to 5%. The *intrinsic energy demand* is then the product of the *energy intensity of GWP* and *GWP*.

[If you happen to model this from scratch, then (1) temporarily assume that the *fraction energy demand to energy supply* is equal to 1, and (2) check whether the model simulates. If you are not modelling from scratch, then just continue with this specification exercise and check your model at the end.]

Sub-model of Fossil Energy Resources

The *production capacity of fossil resources*, initially equal to 415'000'000 bbtu per year, decreases due to *decommissioning of fossil capacity,* which is equal to the *production capacity of fossil resources* divided by the *life expectancy fossil capacity* of 40 years. The *fossil energy production capacity* increases due to *commissioning of new fossil capacity. Commissioning of new fossil capacity* drains the stock of *planned fossil production capacity,* which is initially equal to 60 million bbtu per year. *Planned fossil production capacity* increases through *planning of new fossil capacity*. *Commissioning of new fossil capacity* equals a third-order delay of the *planning of new fossil capacity* with a *planning delay new fossil resources* of 10 years, and with an initial value equal to the *planned fossil production capacity* divided by the *planning delay new fossil resources*.

Planning of new fossil capacity is equal to the *relative capacity extension driven by profitability* as a function of the *relative profitability of fossil resources* (i.e., it is equal to the *relative capacity extension driven by profitability(relative profitability of fossil resources)*), times the *fossil energy production capacity*. Model the *relative capacity extension driven by profitability* with a lookup variable (use a 'lookup' type, not an 'auxiliary

with lookup'). If the input is between 0 and 1 then the output of the lookup is 0% per year, if it is 1.1 then the output is 2% per year, if it is equal to 1.5 then the output is 5% per year, if the input is equal to 2 then the output is 10% per year, if the input is equal to 4 then the output is 15% per year, and if the input is equal to 10 then the output is 18% per year.

The *relative profitability of fossil resources* could be defined as the *average energy price* divided by the *costs of fossil extraction,* which could be assumed to be constant at 5000 dollar per bbtu. The *average energy price* then equals the smallest value of either the *costs of fossil extraction* or the *costs of renewable energy*, times the *price change due to energy shortages.* This variable *price change due to energy shortages* is equal to 0.18 if the *fraction energy demand to energy supply* equals 0.5, to 0.25 if this fraction equals 0.8, to 0.5 if this fraction equal 0.95, to 1.5 if this fraction equals 1.05, to 2 if this fraction equal 1.2, to 2.5 if this fraction equal 1.5, and to 3 if this fraction equals 2. The *fraction energy demand to energy supply* can be defined as the *intrinsic energy demand* divided by the *total energy production*.

Assume for now that the *costs of renewable energy* are equal to the *initial Costs of renewable energy* of 20000 dollar per bbtu. Define for now the *Total energy production* to be equal to the *Fossil energy production capacity*.

[If you happen to model this from scratch, then check whether the model simulates.]

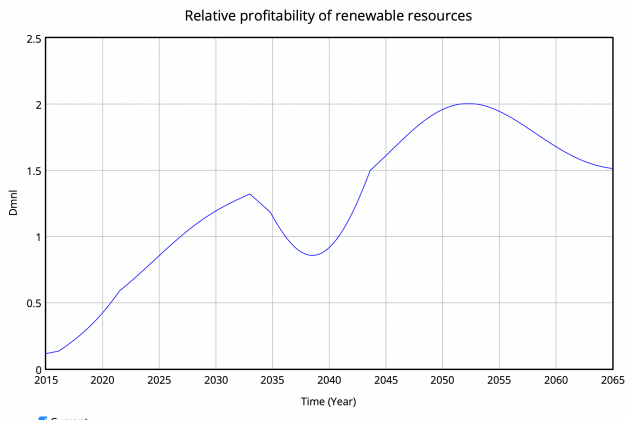
Sub-model of Renewable Resources

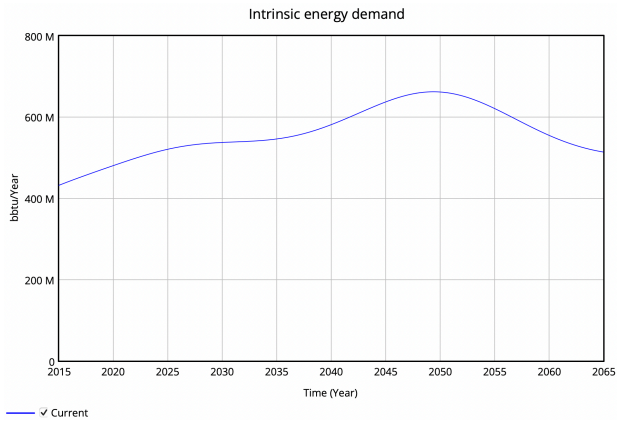
The stock-flow structure of this sub model is basically the same as the one of the fossil capacity. In other words, you should be able to copy-paste the structure and adapt the names and values. Just for the sake of completeness, here's the full description: *renewable energy production capacity*, initially equal to 50'000'000 bbtu per year, decreases due to *decommissioning of renewable capacity,* which is equal to the *renewable energy production capacity* divided by the *life expectancy renewable capacity* of 30 years. *Renewable energy production capacity* increases through *commissioning of new renewable capacity. Commissioning of new renewable capacity* drains the stock of *planned capacity of renewable resources,* which, initially, is equal to 80.8 million bbtu per year. The *planned capacity of renewable resources* increases through *planning of new renewable capacity*. *Commissioning of new renewable capacity* is equal to a third-order delay of the *planning of new renewable capacity* with a *planning delay time new renewable resources* of 10 years. Initially, it is equal to the *planned capacity of renewable resources* divided by the *planning delay time new renewable resources*.

From here on, this sub-model differs from the fossil energy resources sub-model. *Planning of new renewable capacity* equals the sum of the *extension of renewables driven by subsidies* and the *relative capacity extension driven by profitability* as a function of the *relative profitability of renewable resources*, times the *renewable energy production capacity*. In other words, reuse the lookup-variable *relative capacity extension driven by profitability*. Assume the *extension of renewables driven by subsidies* is 10% per year in 2015 and linearly decreases to 0 in 2035, after which it remains 0. The *costs of renewable energy*, initially equal to the aforementioned *initial costs of renewable energy,* decrease via a variable *decreasing costs of renewable energy*. The *decreasing costs of renewable energy* could be specified as the product of the *costs of renewable energy* and the *annual decrease of renewable energy costs* of 7% per year. The *relative*

profitability of renewable resources could be defined as the *average energy price* divided by the *costs of renewable energy*. Model the *relative share renewable resources* as the *renewable energy production capacity* divided by the sum of the *renewable energy production capacity* and the *fossil energy production capacity*.

7. Unfortunately, the model file contains four errors compared to the description above. Correct all errors in the model and define the correct model settings. If you have corrected the model, the simulations should show the following behaviours for the *relative profitability of renewable resources* and *intrinsic energy demand*. Answer the questions below.





   a. (5 / 100) What are the correct model settings for this model? Provide an explanation for your choice of the integration method in one sentence.

   b. (24 / 100) What are the four (structural) errors that found their way into the model, and how do you correct them? List the variable with the error and the corrected equation.

   c. (6 / 100) Conduct a sensitivity analysis in Vensim for the variable *investment delay time energy decoupling,* varying the parameter by +/-10%. Paste (i) a screenshot of the settings of your sensitivity analysis, and (ii) a screenshot of the output graph for the stock *intrinsic energy demand*. (iii) Name the type of sensitivity that you observe. (iv) Provide an explanation based on the model structure, why you observe that sensitivity. (Note: If you didn't manage to debug all errors in the model, you may use the model in the state as it is.)

   d. (5 / 100) What could be improved on the model? Provide two examples of (i) where the model structure falls short and why, (ii) and how you propose to improve the model structure. (Note: An explanation text of the policies is sufficient; you don't have to implement the policies in the model.

ABM modelling questions *(40 points)*

There is a growing scientific literature exploring the equity and justice implications of the energy transition. For example, new electric vehicles where initialize substantially more expensive than equivalent combustion engines. So, they were only adopted by people who could afford them. The same is true for solar PV and nowadays for battery storage. However, at the same time, the various energy transition technologies are characterised by substantial learning effects, causing the costs per unit of performance to rapidly fall. This means that the richer early adopters are paying for this learning which make the technology increasingly affordable for less affluent people. To what extend does this learning effect offset the observed inequality in technology adoption?

The provided ABM is a highly stylized model designed to explore this question. In the model, there are high income and low-income agents. The income of a given agent is drawn from a gamma distribution. At every time step, agents can decide to adopt a novel technology. To adopt the novel technology, they must be able to afford the technology, and they must be willing to adopt the technology.

The affordability question is based on comparing the income of the agent against the cost of the technology. The cost of technology falls, following Wright's law, as a function of the number of adopters. Wright's law means that the cost of a technology falls at a consistent rate as a function of every doubling of the production.

The willingness question is based on a simple Bass diffusion model. The probability of an agent being willing to adopt the technology is based on a based on a base rate of 5% plus 0.02 times the number of other agents that have already adopted the technology. Here a distinction is made between the two income types and an interaction term Ic. In short, if Ic=0, there is no interaction between the two income types, while if Ic=0, the two income types or perfectly mixed.

To start the diffusion process, the model is initialised by randomly giving the technology to 5% of the agents.

8. Implement 3 additional policies, ensuring it is possible to specify which policy is used via a keyword argument when instantiating the model. The strategies to be added are:
    - Seed the technology randomly only in the low-income population
    - Seed the technology to the 5% poorest agents in both income groups
    - Seed the technology to the 10% poorest agents in the low-income group

    a. (8 / 100) Sketch a flowchart or provide in pseudo-code a conceptual description of how to implement the 3 policies

    b. (12 / 100) Implement the 3 policies. For validation purposes, Figure 1 gives you a sense of the dynamics the model produces. The necessary code to create this visual is included in the notebook. Document your answer with a screenshot of the code.

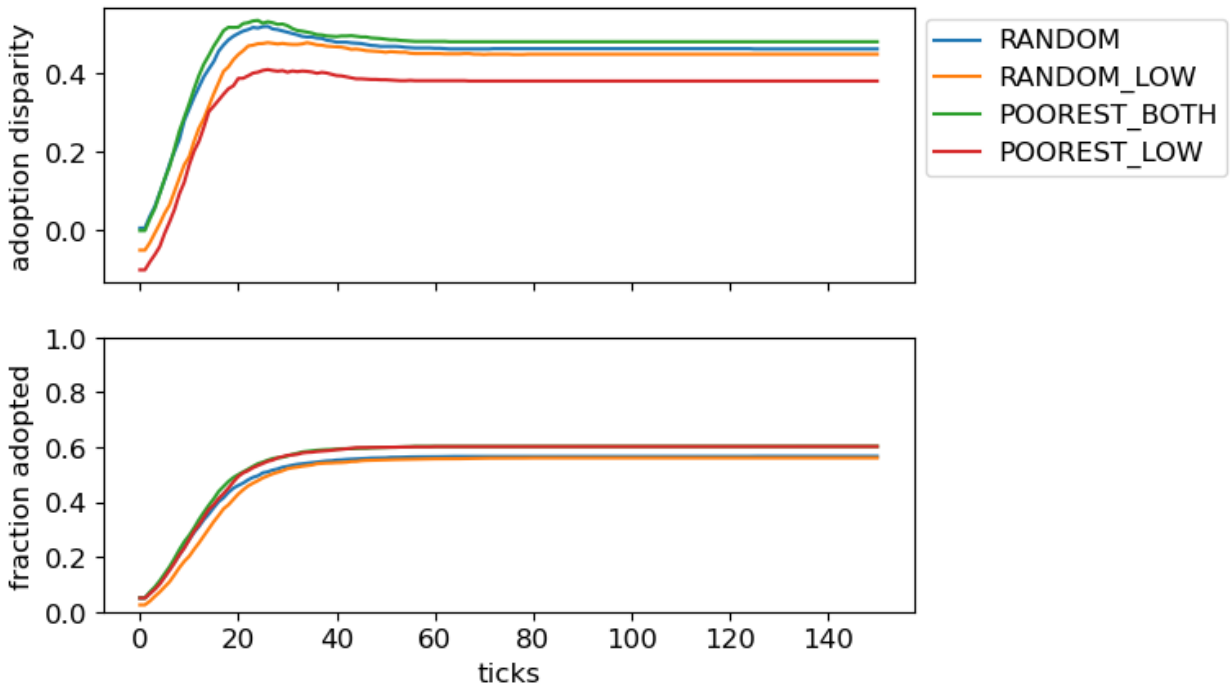


Figure 1 Indicative results for the default model and seed 31415.

9. (8 / 100) The current model is highly simplistic and could trivially be implemented as a system of ordinary equations (which is the source of inspiration for this model). The real strength of ABM is that it can more easily account for social networks and other factors that influence adoption decision.

   Imagine that you are asked to expand the model by explicitly accounting for both the neighbourhood of the agent and its social network. The willingness of adoption then becomes a function of the fraction of adoption in its neighbourhood plus the fraction of adoption in its social network. Pay particular attention to how in the social network you would account for the Interaction Coefficient (Ic).

   Give a conceptual description of any changes in spaces, agent logic, or model initialisation or agent activation logic that would be needed.

10. The model explores the dynamics of adoption of a technology across a high- and low-income agent population. The model is at least behaviourally sensitive to various parameters. Explore the behaviour of the model over the Interaction Coefficient (Ic) and the Learning rate. Vary Ic between 0 and 0.5, Vary the learning rate between 0.15 and 2. For both, use 4 values.

    Please use the originally provided model.

    a. (6 / 100) Implement the code necessary for performing the described set of experiments.
    b. (4 / 100) Provide a clear visualisation of how the model's behaviour changes across the 16 parameter combinations.
    c. (2 / 100) Describe and explain the changes in behaviour as shown in response to question 10b.